1. As width and height ratio is 16:9
   The number of pixel is easy to calculate
   16:9 = $\frac{L}{1125}$, L=2000
   There are 24 pixel exist, as Tv lines interlaced and each field have 1/60 duration, total frame is 30 frame per sec
   So, The total bits would take $Total\ b = (2000 * 1125)pixel * 24(bit) * 60(frame) * 7200 = 11.66 * 10^{12} bit$
   b=11.66Tb

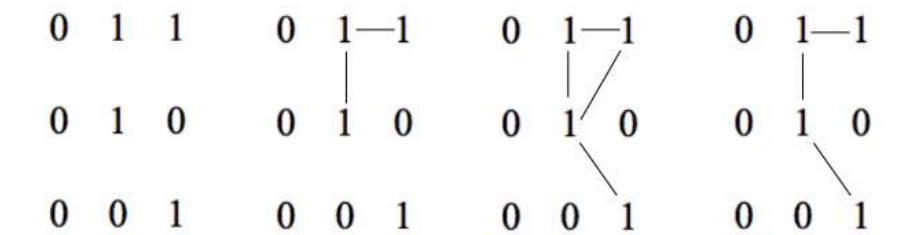
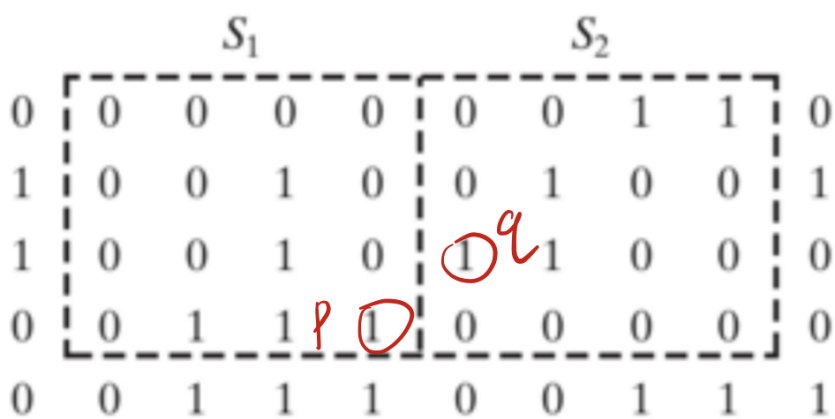


Fig: An arrangement of pixels **Fig: 4-connectivity of pixels** **Fig: 8-connectivity of pixels** **Fig: m-connectivity of pixels**

2. This is the concept of adjacent pixels



   Let's think two pixel p and q
   In this case,
   (a)S1 and S2 are not 4-adjacent becuase q is not in set N4(p)
   (b)S1 and S2 are 8-adjcent because q is the set N8(p)
   (c)S1 and s2 are m-adjacent because q is $N_D(p) and the set N_4 \cap N_4(q) is empty$
   (ambiguity solved)

3. let's integral the function
   $F_1 = \int_0^r 2(1-x)dx \quad F_2 = \int_0^z 2xdx$

$= 2\left(x - \frac{x^2}{2}\right)\Big|_0^r \quad = x^2\Big|_0^z$
$= 2\left(r - \frac{r^2}{2}\right) \quad = z^2$
$F_1 = 2r - r^2$
$F_1 = F_2$
$= z^2 = 2r - r^2$

$$z = \sqrt{2r - r^2}$$

in terms of r and z will accomplish $z = \sqrt{2r - r^2}$

4. The result is different. after taking blur filter, as image1 has more boundary than image2, the result will be different. Code and the histgram is included in problem4.ipynb.

5. Please check histogram.ipynb.

6. Please check gaussian_noise.ipynb.